# A Log-Linear IRT Accumulator Model of Early Word Learning

Draft specification (revised: 2PL extension)

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## 1 Research questions driving the modeling

The model is organized to answer three research questions. The baseline is the minimal specification that lets each be posed and answered; each variant in the nested family (§10) sharpens one of them. A fourth question — when does accumulation start? — is posed in the appendix because the start-time parameter $s$ is near-non-identifiable in practice (see §A).

**RQ1. Is there acceleration in the accumulation process?** McMurray (2007) and Mitchell & McMurray (2009) argued from theory that the empirical "vocabulary explosion" requires acceleration in the threshold distribution $g(\tau)$: pure constant-rate accumulation produces growth that's too smooth, and leveraged learning shifts the curve in time but does not create acceleration that isn't already present. Bergelson (2020) made the empirical case: older infants are "better learners." We formalize this question via $\delta$: in our linear predictor, time enters as $(1+\delta)\log((t-s)/a_0)$, where $\delta = 0$ corresponds to McMurray's pure linear accumulator (one log-token of evidence per log-age unit, no acceleration). Posterior $\delta > 0$ with CrI excluding zero is a quantitative confirmation of the McMurray–Mitchell acceleration claim.

**RQ2. How well does frequency reconstruct word difficulty?** Braginsky et al. (2019) showed frequency is only modestly related to AoA within lexical classes. We test this within our model by asking how much of word-level residual difficulty $\psi_j$ is explained by $\log p_j$, and whether the per-class slope $\beta_c$ on $\log p_j$ deviates from the unit-accumulator baseline $\beta_c = 1$.

**RQ3. How much of child-level outcome variance is input-driven?** This is the $\sigma_\alpha$ question: once we pin the variance of CDS input rate $r_i$ from external data, how much residual child variance must be absorbed by a learning-efficiency term $\alpha_i$? Posterior on $\pi_\alpha = \sigma_\alpha^2/(\sigma_r^2 + \sigma_\alpha^2)$ reports the efficiency share of intercept variance.

Each RQ maps onto a specific posterior quantity, recovered as the nested family is built up:

## 2 One-paragraph summary

A child $i$ of age $a_i$ has, on average, heard word $j$ approximately $T_{ij} = r_i\, p_j\, H\,(a_i - s)$ times, where $r_i$ is the child's tokens-per-hour of CDS, $p_j$ is the word's corpus probability, $H$ is waking hours per month, and $s$ is the global start time for accumulation. The child produces word $j$ on the CDI

| RQ | Quantity | How answered |
|---|---|---|
| 1 | $\delta$ | Posterior in M2 vs. M1; baseline (M1) pins $\delta = 0$. |
| 2 | $\beta_c$, $r(\hat{\psi}_j, \log p_j)$ | Posterior in M4 (per-class $\beta_c$); residual correlation in M3. |
| 3 | $\pi_\alpha = \sigma_\alpha^2/(\sigma_r^2 + \sigma_\alpha^2)$ | Derived posterior in M3+; $\sigma_r^2$ pinned externally. |

Table 1: Mapping from research questions to model quantities and nested-family stages (M1–M5).

with probability $\sigma(\log T_{ij} + \log \alpha_i - \psi_j)$ under a canonical Rasch/IRT logit link ($\sigma(x) = 1/(1 + e^{-x})$), where $\alpha_i$ is the child's learning-efficiency multiplier and $\psi_j$ is the word's log-threshold. Every quantity lives on a log-token scale with $O(1)$ priors, so Stan is well-conditioned. The crucial move is that $r_i$'s population distribution is pinned by external data (Sperry/HR/WF), so $\sigma_\alpha$ is identifiable and Question 4 has an answer. Questions 1–3 are posterior summaries of the baseline; planned variants sharpen them.

# 3 Quantities

| Symbol | Meaning | Units |
|---|---|---|
| $r_i$ | Child $i$'s CDS input rate | tokens / hour |
| $\alpha_i$ | Child $i$'s learning-efficiency multiplier | dimensionless |
| $p_j$ | Word $j$'s corpus probability (CHILDES) | dimensionless, $\sum_j p_j = 1$ |
| $H$ | Waking hours per month (fixed) | hours / month |
| $a_i$ | Age at CDI assessment | months |
| $s$ | Global start time (accumulation onset) | months |
| $\psi_j$ | Log expected tokens of word $j$ to acquisition | log-tokens |
| $\delta$ | Age rate-change exponent | dimensionless |
| $\lambda_j$ | Word $j$'s 2PL discrimination | dimensionless ($> 0$) |
| $a_0$ | Reference age (fixed) | months |
| $y_{ij}$ | Child $i$ produces word $j$ on CDI | $\{0, 1\}$ |

**Fixed constants.** $H = 12 \times 30.44 \approx 365$ hours/month (12 waking hours/day). $a_0 = 20$ months.

# 4 Generative model

## 4.1 Linear predictor

For child $i$, word $j$, age $a_i > s$:

$$\eta_{ij} = \lambda_j \cdot \Big[\log r_i + \log \alpha_i + \log p_j + \log H + (1 + \delta)\, \log\Big(\tfrac{a_i - s}{a_0}\Big) - \psi_j\Big]. \tag{1}$$

Here $\lambda_j \geq 0$ is a word-specific *discrimination* parameter (2PL). With $\lambda_j \equiv 1$ the model collapses to the Rasch form $\eta_{ij} = \log(r_i \alpha_i p_j H(a_i - s)) - \psi_j$ — the log ratio of tokens heard to tokens needed, with unit logit-per-log-token gain. With $\lambda_j$ free, each word has its own gain; high $\lambda_j$ produces a sharp transition near $\psi_j$, low $\lambda_j$ a gradual one. Section 6 motivates the extension in terms of the accumulator and in terms of model fit.

### 4.2 Likelihood

Canonical Rasch/IRT logit link:

$$y_{ij} \mid \eta_{ij} \sim \text{Bernoulli}\big(\sigma(\eta_{ij})\big), \qquad \sigma(x) = \frac{1}{1+e^{-x}}. \tag{2}$$

The logit link is the standard "log-linear IRT" choice and is numerically well-behaved in Stan. $\psi_j$ is interpretable as the log-threshold in log-tokens on the logit scale (the scale factor is absorbed implicitly; $\exp(\psi_j)$ is proportional to, not exactly, the expected number of tokens to 50%-acquisition).

Edge case: when $a_i \le s$, set $\mathbb{P}(y_{ij} = 1) = \varepsilon$ and exclude these cells from the log-age term (in the code this is implemented by flooring $a_i - s$ at 0.01 inside the log).

### 4.3 Priors

**Child-level latent (collapsed).** The likelihood depends on $\log r_i$ and $\log \alpha_i$ only through their sum, so we sample a single child-level latent

$$\xi_i \coloneqq \log r_i + \log \alpha_i \sim \mathcal{N}\big(\mu_r,\ \sigma_r^2 + \sigma_\alpha^2\big).$$

Here $\mu_r = \log 1198$ and $\sigma_r^2 = \text{Var}(\log r)_{\text{Sperry/HR/WF}}$ are *pinned* from external data (this is the identifiability lever for RQ4), while $\sigma_\alpha$ is free:

$$\sigma_\alpha \sim \text{HalfNormal}(0, 1). \tag{3}$$

The per-child $\log \alpha_i$ is a *derived* quantity with closed-form posterior conditional on $\xi_i$:

$$\log \alpha_i \mid \xi_i \ \sim\ \mathcal{N}\left(\frac{\sigma_\alpha^2}{\sigma_r^2 + \sigma_\alpha^2}\,(\xi_i - \mu_r),\ \frac{\sigma_r^2 \sigma_\alpha^2}{\sigma_r^2 + \sigma_\alpha^2}\right),$$

reported in generated quantities. (Rationale for this collapsed form is in §5: sampling both latents individually over-parameterizes the model and creates bad posterior geometry in HMC.)

**Word thresholds by lexical class.**

$$\psi_j \mid \text{class}(j) = c \sim \mathcal{N}(\mu_c, \tau_c^2), \quad \mu_c \sim \mathcal{N}(8, 3^2), \quad \tau_c \sim \text{HalfNormal}(0, 1). \tag{4}$$

**Note on $\mu_c$.** The earlier value $\log 1000$ was a back-of-envelope centering. A more principled choice: tokens heard of a typical word by age 20 mo spans roughly $e^5$ (rare words) to $e^{12}$ (function words), so $\mu_c \sim \mathcal{N}(8, 3^2)$ is weakly informative across the relevant range and dominated by the data.

**Global start time (RQ2).**

$$s \sim \mathcal{N}(4.5, 2^2) \text{ truncated to } [0, 15]. \tag{5}$$

One $s$ for the whole model, per your decision. The prior is centered at 4.5 mo — M&P's comprehension estimate was ∼2 mo; production is likely later. $s$ close to 0 would indicate accumulation begins near birth.

**Age rate-change (RQ3).**

$$\delta \sim \mathcal{N}(0, 0.5^2). \tag{6}$$

**Word discrimination (2PL).**

$$\log \lambda_j \sim \mathcal{N}(0, \sigma_\lambda^2), \quad \sigma_\lambda \sim \text{HalfNormal}(0, 1). \tag{7}$$

Setting $\sigma_\lambda \to 0$ (via a near-zero prior SD passed as data) pins every $\lambda_j = 1$ and recovers the Rasch model. This is the toggle used in code: one Stan file serves both Rasch and 2PL variants. Scale identifiability (the classic $\lambda_j \leftrightarrow 1/\lambda_j$, $\psi_j, \xi_i \leftrightarrow \lambda_j\psi_j, \lambda_j\xi_i$ ambiguity of 2PL) is resolved by our pinning of $\sigma_r$: the $\xi$ scale is fixed externally, so $\psi_j$ and $\lambda_j$ are separately identified.

$\delta = 0$ is the pure cumulative null. $\delta > 0$ indicates older children accumulate faster — a formal test of the Bergelson comprehension-boost claim.

# 5 How each RQ is answered

## RQ1 — is there acceleration?

Single parameter $\delta$, entering the time term as $(1 + \delta) \log((t - s)/a_0)$. $\delta = 0$ is McMurray's pure linear accumulator: each log-age unit contributes one log-token of evidence. $\delta > 0$ means evidence accumulates super-linearly in log-age (or equivalently: tokens grow as $t^{1+\delta}$ on the linear scale, faster than the linear-accumulation baseline). Headline posterior: $\hat{\delta}$ with CrI; CrI excluding zero is a quantitative confirmation of the McMurray–Mitchell argument that acceleration is necessary to produce empirical vocabulary growth.

The natural ablation is `pin_delta=0` (M2 vs. M1 in the nested family, §10), which forces the model back to the pure accumulator. If $\delta$ is doing real work, this ablation should produce a substantially worse fit and force the other parameters into pathological regions.

## RQ2 — does frequency reconstruct word difficulty?

The Rasch baseline (M1 in §10) sets the coefficient on $\log p_j$ to 1 (the pure-accumulator assumption: each token of word $j$ contributes one log-token of evidence). Posterior $\hat{\psi}_j^{(s)}$ is then residual word difficulty after unit-coefficient frequency. We report:

- Posterior correlation $r(\hat{\psi}_j, \log p_j)$ and $R^2$ — what fraction of word-level difficulty is explainable by frequency.
- Posterior class means $\hat{\mu}_c$ — does class structure persist after frequency is accounted for?
- Posterior $\beta_c$ in M4 (per-class slope on $\log p_j$) — do classes differ in how strongly frequency contributes?

The class-specific slope variant M4 is the principal RQ2 test. $\beta_c = 1$ corresponds to the unit-accumulator baseline; $\beta_c < 1$ means frequency contributes less per log-unit; $\beta_c \approx 0$ means frequency is uninformative once per-word $\psi_j$ is free.

## RQ3 — input- vs. efficiency-driven variance

The linear predictor contains $\log r_i + \log \alpha_i$ as a sum, unidentifiable from Wordbank alone. We break the degeneracy by pinning $\sigma_r^2$ at its externally-measured value (from Sperry, Hart & Risley, and Weisleder & Fernald pooled). The child-level latent $\xi_i = \log r_i + \log \alpha_i$ is then identified from the data, with implied distribution $\xi_i \sim \mathcal{N}(\mu_r,\, \sigma_r^2 + \sigma_\alpha^2)$. Subtracting the pinned $\sigma_r^2$ from the inferred $\mathrm{Var}(\xi_i)$ identifies $\sigma_\alpha^2$, giving

$$\pi_\alpha \;\coloneqq\; \frac{\sigma_\alpha^2}{\sigma_r^2 + \sigma_\alpha^2} \tag{8}$$

as a well-defined derived posterior quantity: the proportion of child-level outcome variance attributable to learning efficiency rather than input rate. Its complement, $1 - \pi_\alpha$, is the input-driven fraction.

**What $\pi_\alpha$ does and does not decompose.** The full between-child variance in $\theta_{it}$ at age $t$ is

$$\mathrm{Var}_i(\theta_{it}) = \sigma_\xi^2 + \log_{age}^2\, \sigma_\zeta^2 + 2\, \rho_{\xi\zeta}\, \sigma_\xi\, \sigma_\zeta\, \log_{age}\,.$$

At $t = a_0$ (the pivot, where $\log_{age} = 0$), this collapses to $\sigma_\xi^2$ and $\pi_\alpha$ is exactly the efficiency share. Off-pivot, the per-child slope variance $\sigma_\zeta^2$ contributes increasingly. With our fitted $\sigma_\zeta \approx 3.5$ on English, at $\log_{age} = -1$ (e.g., 7 mo when $a_0 = 19$) the slope component contributes $\sim 5\times$ the intercept variance.

We do not decompose $\sigma_\zeta$ into input-rate-growth and efficiency-growth components, because the model treats $\log r_i$ as age-invariant (see "Age-invariance of $\log r_i$" in § Assumptions). $\zeta_i$ therefore acts uniformly on $(\log r_i + \log \alpha_i)$ — without longitudinal observations of $r_i(t)$ within child, the data cannot separate it. We report $\pi_\alpha$ as the intercept-share at the pivot age, and treat $\sigma_\zeta$ as a third axis of between-child variation that exists alongside the input/efficiency intercept decomposition.

The `drop_slopes` ablation ($\sigma_\zeta = 0$) tests whether the intercept-level $\pi_\alpha$ is robust: forced into a single-locus model, the data still gives $\pi_\alpha \approx 0.92$ on English (vs. 0.91 with slopes free), so the headline number is robust to slope-vs-no-slope choices.

**Why the collapsed parameterization matters practically.** Sampling $\log r_i$ and $\log \alpha_i$ individually — each with its own non-centered $\mathcal{N}(0, 1)$ prior and a scale parameter — leaves the posterior with a flat ridge: any rotation of the pair that preserves $\xi_i$ has the same likelihood. Only the priors break the degeneracy, at the cost of 2I parameters where I are identified. In HMC this manifests as chains getting stuck in bad regions (we observed one chain running 28+ minutes on a sim the others cleared in 3). Sampling $\xi_i$ directly, with $\sigma_\alpha$ entering only through $\sigma_{\xi_i}^2 = \sigma_r^2 + \sigma_\alpha^2$, eliminates this geometry entirely. Parameter recovery on a 250-child $\times$ 150-word synthetic dataset: $\pi_\alpha$ posterior 0.62 [0.50, 0.72] against true 0.61.

**Sensitivity analysis.** Refit with $\sigma_r^2 \in \{0.5, 1, 2\} \cdot \sigma_r^{2,\mathrm{obs}}$. Stability of $\pi_\alpha$ across these indicates how much the claim depends on trusting the external estimate.

## 6 Why 2PL: one-token-one-drop is too strong

The Rasch version of (1) (i.e. with $\lambda_j \equiv 1$) implies every word has the same logit-per-log-token gain. For a fixed word, the predicted slope of $\operatorname{logit} p_{ij}$ with respect to age is $(1+\delta)/(a_i - s)$ — on the order of 0.08 logits per month at typical observed ages. Real CDI words transition from ∼5% to ∼95% acquired over roughly six months (about 6 logits), which is ∼12× faster than the Rasch slope can produce.

When we fit Rasch to Wordbank (500 children × 200 items), the posterior finds this mismatch in a predictable way: it cranks every available steepening lever — $\delta$ rises to $\sim 2.3$, $s$ pushes to $\sim$ 12 months, and crucially $\sigma_\alpha$ inflates to $\sim 2$ — implying children differ by $\sim 50\times$ in learning efficiency, and $\pi_\alpha \approx 0.93$. This pattern is stable across the s/$\delta$ identifiability, suggesting misspecification rather than genuine individual-difference variance.

**The 2PL fix.** Adding word-specific discrimination $\lambda_j$ gives each word its own slope around threshold: the per-word logit derivative becomes $\lambda_j(1+\delta)/(a_i - s)$. Words whose referents are highly diagnostic of their meaning ("ball" when you see a ball; concrete nouns generally) can have large $\lambda_j$ and transition sharply; words whose meaning is more distributional or context-dependent (function words, abstracts) have small $\lambda_j$ and gradual transitions. This is the standard two-parameter logistic IRT extension and is strongly preferred over 1PL for CDI data in prior work (Kachergis, Marchman, Dale, Mankewitz, & Frank, 2022, *JSLHR*).

**Three accumulator interpretations of $\lambda_j$ (all observationally equivalent from CDI alone).**

1. **Information per token.** $\lambda_j$ is the per-token shift in the child's certainty about word $j$. High $\lambda_j$: each exposure is diagnostic.
2. **Sharpness of the cognitive threshold.** $\lambda_j$ is the inverse width of the band over which evidence must accumulate for acquisition to occur.
3. **ELI-to-token ratio (bridge to Mollica & Piantadosi).** If only a fraction of heard tokens are effective learning instances and the rest are background noise, $\lambda_j$ is proportional to that fraction for word $j$.

**Falsification checks.** The hierarchical prior on $\log \lambda_j$ (pooled via $\sigma_\lambda$) regularizes: one unusual word can't take over. Two further checks protect against freely flexible fit-to-noise:

1. $\sigma_\alpha$ should *decrease substantially* under 2PL. If the Rasch inflation was misspecification dressed as person variance, 2PL should absorb it into $\lambda_j$ and drop $\sigma_\alpha$ toward a more reasonable value (e.g. ∼0.7), sharpening our answer to RQ4.
2. $\lambda_j$ should correlate with measurable word properties: higher for concrete nouns, lower for function words and abstract adjectives. If $\lambda_j$ is uncorrelated with any external measurement, we have over-fit rather than learned.

**What this changes for the RQs.** RQ1 (frequency-vs-difficulty) unchanged: we still report the posterior correlation of $\psi_j$ with $\log p_j$. RQ2 (start time), RQ3 ($\delta$) unchanged at the

structural level, but the extreme posteriors observed under Rasch should moderate. RQ4 (variance decomposition) sharpens substantially: $\pi_\alpha$ becomes the proportion of child variance after item-level discrimination has been accounted for, which is the quantity we really want.

# 7 Assumptions, listed

1. **Pure accumulation.** Exposure accumulates linearly at a per-token rate independent of prior knowledge. No leveraged learning (Mitchell & McMurray 2009 showed leverage cannot *create* acceleration that is not already present).

2. **Per-word evidence gain $\lambda_j$.** In 2PL mode each token contributes $\lambda_j$ units of logit-scale evidence; $\lambda_j = 1$ reduces to the Rasch "one token, one drop" rule. Per-child gain is *not* modeled: all children receive the same $\lambda_j$ for a given word. This is relaxable by adding a random slope, which we defer.

3. **Logit link with fixed scale.** Canonical Rasch/2PL IRT. Equivalent (up to a $\sim 1.7$ scale factor) to a probit model; approximates the Gamma cumulative of Mollica & Piantadosi for typical parameter values.

4. **Cross-sectional data.** Each child observed once.

5. **CDS only.** Overheard speech ignored in v1.

6. **Same input distribution as reference studies.** Our CDI sample has the Sperry/HR/WF distribution of $r_i$; SES moderators not modeled in v1.

7. **Age-invariance of $\log r_i$.** The model treats each child's input rate as a single per-kid constant, not a function of age. The per-child age-acceleration $(1 + \delta + \zeta_i)$ acts on the combined product $r_i \cdot \alpha_i$, so any age-varying child-side dynamics (efficiency growth, parents talking more to older kids) are absorbed into $\delta+\zeta_i$ without attribution. Empirically supported by within-kid regressions of $\log r_{iv}^{\text{obs}}$ on age in BabyView ($N = 20$ kids, 5688 videos: mean within-kid slope $-0.006$ logits/mo, median $-0.002$) and SEEDLingS ($N = 44$ kids, 525 LENA recordings: mean $-0.011$ logits/mo). Both are at least an order of magnitude smaller than the $(1 + \delta)\ \log_{age}$ effect over the same age range, so $\sigma_\zeta$ is unlikely to be primarily absorbing input-rate growth. See `model/scripts/input_rate_vs_age.R`.

8. **Global $s$.** One start time for all words. Per-class or per-word $s$ is a planned extension.

9. **CHILDES frequencies as truth.** Sampling noise in $p_j$ ignored; defensible for common words.

10. **Lexical class only via thresholds.** Class enters through the hierarchical prior on $\psi_j$; class-specific frequency slopes or start times are variants, not baseline.

# 8 Sampling strategy: children and items

The data we fit is large enough that we always work on a subsample. The strategy is the same across pipelines (Wordbank longitudinal, BabyView, future Seedlings) so results are directly comparable.

**Children.**

1. Filter to children with $\geq 2$ admins (longitudinal coverage).
2. Compute each child's *median admin age* as their representative age.
3. Bin children into $N_{\text{age}} = 4$ age quantile bins.
4. Within each bin, take the top kids by number of admins, then by age span. (Tie-breaking on richer longitudinal data.)
5. Top up uniformly at random if any bin is short (rare).

This replaces an earlier, biased "top-by-priority" rule that just maximized $n_{\text{admin}} + \text{span}/5$; that rule overweighted data-rich kids in narrow age windows. The age-stratified rule guarantees coverage across the whole 16–30 mo developmental window.

**Items.**

1. Filter to items with non-zero CHILDES probability $p_j$.
2. For each lexical class, bin items into $N_{\text{diff}} = 4$ difficulty quartiles based on $\log p_j$.
3. Sample uniformly within each (class, difficulty-quartile) cell — 16 cells for 4 classes $\times$ 4 quartiles.
4. Top up at random if any cell is short.

The earlier item-sampling rule stratified only on lexical class, which oversampled high-frequency words (those are dense in the CHILDES distribution). The class-by-difficulty grid ensures we have hard, medium, and easy words within *each* class, which sharpens the identification of $\psi_j$ and $\beta_c$ when they enter as parameters.

**Pipeline-shared item pool.** All Wordbank-based pipelines (cross-sectional and longitudinal, English and Norwegian) draw from the *union of WG and WS forms*, mirroring the BabyView treatment. Each (child, age, form) is its own admin; a child who took both WG and WS at different ages contributes both. This extends English age coverage from the WS-only 16–30 mo to 8–30 mo and matches the cross-pipeline data definition.

**Per-pipeline application.**

- **Wordbank longitudinal (English / Norwegian).** 200 children $\times$ 200 items per dataset, drawn via the stratified sampling above. English: median 3 admins, 14 mo span; Norwegian: median 8 admins, 14 mo span (Norwegian has the much denser Kristoffersen contributor).
- **Wordbank cross-sectional (English).** 500 children, treated as the most recent (child, form) admin from `long_items.rds`. Stratified the same way as the longitudinal pipeline.
- **BabyView.** 20 longitudinal English subjects (the entire available pool) $\times$ 200 items, with input rate observed via 5 688 videos. No child-side subsampling: we use everyone.

## 9 What we deliver

1. **RQ-addressed posteriors**: $\delta$ (RQ1), $\beta_c$ and $r(\hat{\psi}_j, \log p_j)$ (RQ2), and $\pi_\alpha$ (RQ3) — all with credible intervals.
2. **Per-word** $\psi_j$: absolute log-token thresholds for all 680 CDI items.
3. **Per-child** $\log \alpha_i$: child-level efficiency estimates.
4. **Counterfactual simulator**: given hypothetical $(r, \alpha, a)$, predicted vocabulary size with CrI.
5. **Planned model comparisons**: LOO-CV across the baseline plus variants for each RQ (see §8).

## 10 Nested model family

We build the model as a strictly nested progression from the simplest possible per-word IRT (M0) up to the full-feature variant. Each step adds one named component and answers one specific question; comparing adjacent models (LOO-CV / WAIC, posterior shifts on the structural parameters) attributes the marginal value of that component.

The model classes $c \in \{1, \ldots, C\}$ of word $j$ (lexical category: nouns / predicates / function words / other) enter only through hierarchical priors on $\psi_j$ (class-mean $\mu_c$, class-SD $\tau_c$) and, in M4, through class-specific frequency slopes $\beta_c$. Throughout, $\xi_i = \log r_i + \log \alpha_i$ is the per-child intercept with prior $\xi_i \sim \mathcal{N}(\mu_r, \sigma_r^2 + \sigma_\alpha^2)$ and $\sigma_r$ externally pinned.

**M0 — minimal IRT (no time, no frequency).**

$$\eta_{ijt} = \xi_i + \log H - \psi_j.$$

Per-word $\psi_j$ absorbs all between-item variance; per-child $\xi_i$ absorbs all between-child intercept variance. No age effect, no frequency. This is the absolute null: vocabulary trajectories are flat in age, and word difficulty is a free per-word parameter only.

**M1 — McMurray null (add unit-coefficient frequency and unit-rate accumulation).**

$$\eta_{ijt} = \xi_i + \log \tfrac{t-s}{a_0} + \log H + \log p_j - \psi_j.$$

Adds the structural accumulator content: frequency enters with unit coefficient (one log-token of evidence per token of word $j$ heard), and the time term grows at unit rate per log-age. This is McMurray (2007)'s pure linear accumulator. The slope on $\log_{age}$ is fixed at 1, with no per-child deviation. **M1 vs. M0 quantifies whether frequency and time contribute beyond per-word $\psi_j$ and per-child $\xi_i$** (RQ2 partial answer).

**M2 — add population acceleration $\delta$ (RQ1).**

$$\eta_{ijt} = \xi_i + (1+\delta)\, \log \tfrac{t-s}{a_0} + \log H + \log p_j - \psi_j, \quad \delta \sim \mathcal{N}(0,\, 0.5^2).$$

Frees the $\log_{age}$ slope as a deviation $\delta$ above the pure-accumulator baseline. $\delta = 0$ recovers M1. **M2 vs. M1 is the RQ1 test**: does the data require super-linear-in-age token accumulation?

**M3 — add per-child slope deviations $\zeta_i$.**

$$\eta_{ijt} = \xi_i + (1 + \delta + \zeta_i)\, \log\tfrac{t-s}{a_0} + \log H + \log p_j - \psi_j.$$

Each child has their own deviation $\zeta_i$ from the population slope. Joint per-child prior: $(\xi_i, \zeta_i) \sim \mathrm{MVN}(0, \Sigma)$ with LKJ on correlation; sum-to-zero centering separates $\delta$ from $\mathrm{mean}(\zeta_i)$. Requires longitudinal data for $\sigma_\zeta$ to be identified. **M3 vs. M2 measures between-child variation in growth rate.**

**M4 — add class-specific frequency slope $\beta_c$ (RQ2).**

$$\eta_{ijt} = \xi_i + (1 + \delta + \zeta_i)\, \log\tfrac{t-s}{a_0} + \log H + \beta_{c(j)}\, \log p_j - \psi_j, \quad \beta_c \sim \mathcal{N}(1,\, 0.5^2).$$

Frequency now enters with class-specific weight. $\beta_c = 1$ is M3 (unit accumulator); $\beta_c = 0$ means frequency is uninformative for class $c$ once per-word $\psi_j$ is free. **M4 vs. M3 is the RQ2 test**: do classes differ in how strongly frequency contributes?

**M5 — add per-word discrimination $\lambda_j$ (2PL).**

$$\eta_{ijt} = \lambda_j \Big[\xi_i + (1 + \delta + \zeta_i)\, \log\tfrac{t-s}{a_0} + \log H + \beta_{c(j)}\, \log p_j - \psi_j\Big], \quad \log \lambda_j \sim \mathcal{N}(0,\, \sigma_\lambda^2).$$

Per-word multiplicative scaling on the $(\theta - \beta)$ gap. $\sigma_\lambda \to 0$ recovers M4 (Rasch). **M5 vs. M4 measures item-level sharpness variance.**

**Robustness variants (off the main spine).**

| Variant | Compared to | What it tests |
|---|---|---|
| `lmm` | M3 | Linear-in-age ($(\beta_{age} + \zeta_i)(t - a_0)$) instead of log-linear; tests functional-form sensitivity. |
| `no_class` | M3 | Drops class hierarchy on $\psi_j$; tests whether class structure adds anything. |
| `free_s` | M3 | Frees $s$; tests $s/\delta$ trade-off (see Appendix A). |
| `no_freq` | M2 | Drops $\log p_j$; tests whether frequency adds beyond per-word $\psi_j$. |

**The variant grammar implements the family.** `variant_hyperpriors()` in `model/R/helpers.R` maps a variant name onto the prior overrides that select a specific model in the family. Tight priors near zero pin a parameter off (e.g., $\sigma_\zeta \sim \mathcal{N}(0, 0.001)$ pins $\zeta_i \equiv 0$); loose priors free it. The running log of fits is in `notes/experiments.md`.

## 11 Datasets and observation channels

The variant grammar (§10) toggles *model components*. Independent of that, each dataset comes with its own *observation channels* that determine which auxiliary parameters are appropriate. We separate the two levers in code so they can't be mixed by accident.

**Lever 1 — variant grammar (per-fit toggles).** Tight-prior switches on $\sigma_\lambda$, $\sigma_\zeta$, $s$, $\delta$. Implemented in `variant_hyperpriors()` (`model/R/helpers.R`); identical across pipelines (the `long_`, `io_` prefixes are stripped inside).

**Lever 2 — dataset-level design choices (per-bundle constants).** Properties inherent to the observation channel of a given dataset. Baked into `stan_data` by the `prepare_*.R` script; not a choice a researcher makes per fit. Two examples currently:

- *Reactivity inflation $\beta_{react}$.* Per-child observed input rate $\log r_{iv}^{\text{obs}}$ for video $v$ of child $i$ is $\log r_i + \beta_{\text{react}} + \varepsilon_{iv}$. The bias $\beta_{\text{react}}$ absorbs systematic differences between observed input and "true" input.
  - **BabyView** (head-mounted video, on-camera observer effect): $\beta_{\text{react}} \sim \mathcal{N}(0.4, 0.4)$ — active, with the prior centered around $\sim$1.5x inflation.
  - **SEEDLingS** (LENA all-day audio, no on-camera observer): $\beta_{\text{react}} \sim \mathcal{N}(0, 0.001)$ — pinned at zero. Passive-recording channel, so no inflation.
- *Per-recording weighting.* Currently uniform; planned hook in `stan_data$log_r_obs_weight` for downweighting short or contaminated recordings. Off by default.

**Stan model files form a partial order.** Each file *adds* components on top of the previous, all toggleable via tight priors:

| File | Adds | Used by |
|---|---|---|
| `log_irt.stan` | cross-sectional baseline $\xi_i$ | Wordbank cross-sectional fits |
| `log_irt_long.stan` | admin→child indexing, $\zeta_i$, bivariate prior | Wordbank longitudinal fits |
| `log_irt_io.stan` | per-recording $\log r^{\text{obs}}$ measurement layer, $\beta_{\text{react}}$ | BabyView, SEEDLingS |
| `log_irt_proc.stan` *(planned)* | noisy-$\alpha$ observation from LWL RT/accuracy | Peekbank-linked Stanford datasets |

The result is a small partial order of model files, not a combinatorial explosion. Adding a new dataset usually means writing a `prepare_<dataset>.R` script that reuses an existing Stan file with appropriate dataset-level constants; only when a genuinely new *observation channel* appears do we add a Stan file.

# A Start time $s$ and the $s/\delta$ ridge

A natural fourth question — *when* does accumulation start? — is deferred to this appendix because the parameter $s$ is near-non-identifiable in practice. Mollica & Piantadosi (2017) estimated $s \approx 2$ months for comprehension; we attempted a similar estimate for production but found the data does not support it, for the reason described below.

**Identifiability in principle vs. practice.** When $s$ is freed jointly with $\delta$, the two parameters trade off along a near-perfect ridge: posterior $r(s, \delta) \approx -0.96$ on both English and Norwegian longitudinal fits. In principle the pair is identifiable: setting $(1+\delta)\log(t-s) = (1+\delta')\log(t-s')$ for all $t$ and differentiating in $t$ gives $(1+\delta)/(t-s) = (1+\delta')/(t-s')$, forcing $\delta = \delta'$ and $s = s'$. Practically the curve-shape difference between two points along the ridge is small. For the English age range (16–30 mo), the ridge endpoints $(s, \delta) = (0.5, 9.4)$ and $(3, 8)$ produce $(1+\delta)\log((t-s)/a_0)$ curves that differ by only $\sim 0.1$ logits in shape (after subtracting a constant that $\xi_i$ absorbs). With Bernoulli observations and finite $N$, that shape difference is at the noise floor. So $s$ is *near-non-identifiable in practice*, and the reported $s$ posterior in any `free_s` variant is only a coordinate choice along the ridge — it does not deliver an interpretable estimate of

"onset of accumulation." We pin $s \approx 0$ in M1–M5 and treat `free_s` as a robustness check, not a first-class estimate.

**Reference age $a_0$ is dataset-specific.** $a_0$ is the pivot at which $\log_{age} = 0$ and so $\theta_{it} = \xi_i$ (per-child intercept). Setting a single global $a_0 = 20$ across datasets makes $\xi_i$ language-relative in a misleading way: when admin ages skew far from $a_0$, the per-kid likelihood ridge in $(\xi_i, \zeta_i)$ is tilted by an amount proportional to $\log(\bar{t}_i/a_0)$, pulling the marginal $r(\xi, \zeta)$ across kids in one direction or the other. We observed this as a spurious sign flip in $\rho_{\xi\zeta}$ between English (median admin age 19; $r = +0.39$) and Norwegian (median admin age 26; $r = -0.32$) on the same model. Re-evaluating each kid's $\xi$ at their personal mid-observation age makes the flip vanish (Norwegian $r$ becomes $+0.08$). We therefore set $a_0$ per dataset to the median admin age, in each `prepare_*.R` script. This is a reparameterization, not a model change: the likelihood, $\delta$, and $\zeta_i$ are invariant; only the *interpretation* of $\xi_i$ changes (from "ability at 20 mo" to "ability at the dataset's median admin age").